Two asymptotic formulas for square Hardinian arrays

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Abstract

We prove that the numbers of square Hardinian arrays with parameters $r = 2$ and $r = 3$ satisfy $H_2(n,n) \sim 16^n/(324\pi n)$ and $H_3(n,n) \sim 64^n/(78732\pi n^3)$, respectively. These are two of the asymptotic formulas recorded as conjectures by Dougherty-Bliss and Kauers. Starting from their complementary-minor counting identity, we obtain exact sums of squares of binomial sums. A uniform local expansion, together with a geometric tail bound, reduces the leading constants to two small moment determinants. The argument treats fixed $r = 2, 3$ as $n \to \infty$ and does not use a conjectured recurrence.

1 Statement and source identity

Following Dougherty-Bliss and Kauers [1], let $H_r(n,n)$ be the number of arrays $(a_{i,j})_{1\le i,j\le n}$ of nonnegative integers such that

1. $a_{1,1} = 0$ and $a_{n,n} = n - r - 1$;
2. every step one position to the right, down, or diagonally down and right increases the entry by either 0 or 1;
3. $|a_{i,j} - (\max(i,j) - 1)| \le r$ for all i, j.

The step condition is imposed whenever both positions belong to the array. We consider $r \in \{2, 3\}$ fixed.

Theorem 1. *As $n \to \infty$ through the positive integers,*

$$H_2(n,n) \sim \frac{16^n}{324\pi n}, \qquad H_3(n,n) \sim \frac{64^n}{78732\pi n^3}.$$

The two formulas occur in the conjectural asymptotic table on page 12 of the published version of [1]. OEIS A253217 credits the $r = 2$ asymptotic conjecture to Vaclav Kotesovec on March 2, 2023 [2]. Our starting point is the counting identity displayed in the proof of its Theorem 6, on page 9; we use the identity itself, rather than its consequence of D-finiteness. Put $N = n - 2$ and let

$$M = \left[\binom{i+j}{i}\right]_{0\le i,j\le N}.$$

For $n \ge r + 1$, that identity reads

$$H_r(n,n) = \sum_{\substack{I,J\subseteq\{0,\dots,N\}\\ |I|=|J|=r}} \det M_{I^c,J^c}. \tag{1}$$

Rows and columns of every submatrix are placed in increasing order; complements are taken in $\{0,\dots,N\}$ and the empty determinant is 1. Thus the matrix has $n-1$ rows and columns, and exactly r of each are deleted. In particular, each subset occurs once, with no factorial factor. At $n=r+1$, both the array definition and (1) give 1.

We derive the theorem from (1). We make no claim here about the asymptotic formulas for general r, or about completion of an exhaustive priority search.

2 A sum of squares

Let $P_{u,i}=\binom{u}{i}$ for $0\le u,i\le N$, with a binomial coefficient equal to zero outside its usual range. The Vandermonde identity gives $M=PP^T$. The matrix P is unit lower triangular, and its inverse has entries

$$(P^{-1})_{u,i}=(-1)^{u+i}\binom{u}{i}.$$

Writing $E=\operatorname{diag}((-1)^0,\dots,(-1)^N)$ and $B=P^TP$, we therefore have $\det M=1$ and $M^{-1}=EBE$. Jacobi's complementary-minor identity now gives

$$\det M_{I^c,J^c}=(-1)^{\sum I+\sum J}\det(M^{-1})_{J,I}=\det B_{J,I}.$$

The last equality uses the identical sign supplied by the two diagonal matrices E. Applying Cauchy–Binet to $B=P^TP$ and summing over I,J in (1) yields

$$H_r(n,n)=\sum_{0\le u_1<\cdots<u_r\le N} W_r(u_1,\dots,u_r)^2, \tag{2}$$

where

$$W_r(u_1,\dots,u_r)=\sum_{0\le i_1<\cdots<i_r\le N}\det\left[\binom{u_a}{i_b}\right]_{a,b=1}^{r}. \tag{3}$$

Indeed, for each row set U, the sums over I and J are the same sum of minors, so their product is its square. All sums in this step are finite.

For independent random variables $X_a\sim\mathrm{Bin}(u_a,1/2)$, define

$$w_r(u_1,\dots,u_r)=2^{-\sum_a u_a}W_r(u_1,\dots,u_r).$$

Expanding the determinant in (3) and sorting the distinct values of the X_a shows that

$$w_r(u_1,\dots,u_r)=\mathbb{E}\left[\operatorname{sgn}\prod_{a<b}(X_b-X_a)\right]. \tag{4}$$

For distinct values, the determinant sign is exactly the sign of the sorting permutation. A tie contributes zero on the right and is absent from the strictly increasing column sets on the left. Here $\operatorname{sgn}(0)=0$. In particular, we have the global bound

$$|w_r(u_1,\dots,u_r)|\le 1. \tag{5}$$

3 Exact formulas for two and three rows

For integers $0 \le u < v$, set

$$D(u,v) = \sum_{j=u}^{v-1} \binom{u+v}{j}, \qquad Q(u,v) = 2^{-u-v} D(u,v). \tag{6}$$

If $X \sim \mathrm{Bin}(u, 1/2)$ and $Y \sim \mathrm{Bin}(v, 1/2)$ are independent, then $Y - X$ has the distribution of $Z - u$ with $Z \sim \mathrm{Bin}(u+v, 1/2)$. By symmetry, $\mathbb{P}(Z > u) = \mathbb{P}(Z < v)$, and hence

$$\mathbb{P}(X < Y) - \mathbb{P}(X > Y) = \mathbb{P}(u \le Z < v) = Q(u,v).$$

It follows from (4) that $W_2(u,v) = D(u,v)$ and

$$H_2(n,n) = \sum_{0 \le u < v \le N} D(u,v)^2. \tag{7}$$

For any three real numbers, including ties, the identity

$$\mathrm{sgn}((y-x)(z-x)(z-y)) = \mathrm{sgn}(y-x) - \mathrm{sgn}(z-x) + \mathrm{sgn}(z-y) \tag{8}$$

follows by checking their relative order. Taking expectations gives

$$w_3(u,v,w) = Q(u,v) - Q(u,w) + Q(v,w) \quad (0 \le u < v < w). \tag{9}$$

Thus, if

$$A(u,v,w) = 2^w D(u,v) - 2^v D(u,w) + 2^u D(v,w),$$

then $W_3(u,v,w) = A(u,v,w)$ and

$$H_3(n,n) = \sum_{0 \le u < v < w \le N} A(u,v,w)^2. \tag{10}$$

Equations (7) and (10) are exact integer identities.

4 A uniform local expansion

Write $u = N - a$, $v = N - b$, where $0 \le b < a \le L$ and $L = O(\log N)$, and put $d = a - b$, $s = a + b$.

Lemma 2. *Uniformly for these offsets,*

$$Q(N-a, N-b) = \frac{d}{\sqrt{\pi N}} \left[1 + \frac{s/4 - d^2/12 - 7/24}{N} \right] + O\left(\frac{(1+L)^5}{N^{5/2}} \right). \tag{11}$$

Proof. Put $m = 2N - s$. The summation indices in (6) can be written $j = m/2 + t$, with $t = -d/2, -d/2+1, \ldots, d/2-1$. The numbers t may be half-integers, but j and $m - j$ are integers. The Stirling expansion with remainder gives, uniformly here,

$$2^{-m} \binom{m}{m/2+t} = \sqrt{\frac{2}{\pi m}} \left[1 - \frac{2t^2 + 1/4}{m} + O\left(\frac{(1+|t|)^4}{m^2} \right) \right]. \tag{12}$$

For completeness, set $x = 2t/m$. Applying the logarithmic Stirling formula to $m!$, $(m/2+t)!$, and $(m/2-t)!$ gives

$$\log\left(2^{-m}\binom{m}{m/2+t}\sqrt{\frac{\pi m}{2}}\right) = -\frac{m}{2}\left[(1+x)\log(1+x)+(1-x)\log(1-x)\right] \\ -\frac{1}{2}\log(1-x^2)+\frac{1}{12m}-\frac{1}{12(m/2+t)}-\frac{1}{12(m/2-t)}+O(m^{-3}).$$

Since $|t| = O(\log N)$ and $m \asymp N$, the three factorial arguments are uniformly comparable to N, and eventually $|x| \le 1/2$. Taylor expansion therefore makes the last display equal to

$$-\frac{2t^2+1/4}{m}+O\left(\frac{t^2}{m^2}+\frac{t^4+1}{m^3}\right).$$

Exponentiating proves (12), with an error constant uniform over the logarithmic window. Substituting $m = 2N - s$ into (12) gives each mass as

$$\frac{1}{\sqrt{\pi N}}\left[1+\frac{s/4-t^2-1/8}{N}+O\left(\frac{(1+L)^4}{N^2}\right)\right].$$

Finally,

$$\sum_{k=0}^{d-1}(k-d/2)^2 = \frac{d(d^2+2)}{12}.$$

There are $d \le L$ terms, so summing yields (11). □

For $r = 2$, the lemma immediately implies

$$w_2(N-a, N-b) = \frac{a-b}{\sqrt{\pi N}}(1+o(1)) \tag{13}$$

uniformly in the window. For $r = 3$, take $a > b > c \ge 0$ and apply the lemma to the three terms in (9). The coefficients of $N^{-1/2}$ cancel because $(a-b)-(a-c)+(b-c) = 0$. In the next coefficient, the terms involving $ds = a^2 - b^2$ and the constant multiple of d also cancel. Since

$$(a-b)^3-(a-c)^3+(b-c)^3 = -3(a-b)(a-c)(b-c),$$

we obtain, with $\Delta(a,b,c) = (a-b)(a-c)(b-c)$,

$$w_3(N-a, N-b, N-c) = \frac{\Delta(a,b,c)}{4\sqrt{\pi}\,N^{3/2}}+O\left(\frac{(1+L)^5}{N^{5/2}}\right). \tag{14}$$

For distinct integer offsets $\Delta(a,b,c) \ge 2$, so the relative error in (14) tends uniformly to zero. The same is true in (13), since $a-b \ge 1$. The next-order term in the local binomial expansion is essential in the three-row case.

5 Tail control and evaluation of the constants

Let $r \in \{2,3\}$ and $R = \binom{r}{2}$. Replacing the row indices in (2) by decreasing offsets gives

$$\frac{N^R H_r(n,n)}{2^{2rN}} = \sum_{N \ge a_1 > \cdots > a_r \ge 0} 4^{-\sum_i a_i} N^R w_r(N-a_1, \ldots, N-a_r)^2. \tag{15}$$

Choose $L = \lceil 6 \log_4 N \rceil$. By (5), the part of the sum with $a_1 > L$ is at most

$$N^R \sum_{\max_i a_i > L} 4^{-\sum_i a_i} \le N^R r \left(\sum_{k=L+1}^{\infty} 4^{-k} \right) \left(\sum_{k=0}^{\infty} 4^{-k} \right)^{r-1}$$
$$= \frac{r}{3} \left(\frac{4}{3} \right)^{r-1} N^R 4^{-L} = O(N^{R-6}) = o(1). \tag{16}$$

For this upper bound we have enlarged the sum to all nonnegative integer tuples.

Inside the window, (13) and (14) have uniform relative errors. A squared Vandermonde is a fixed polynomial in the offsets, so its sum against $4^{-\sum_i a_i}$ converges. Consequently the error in the weighted window sum tends to zero, while the truncated main sum tends to the full convergent sum. Combining this fact with (16) proves

$$H_2(n,n) \sim \frac{16^N}{\pi N} Z_2(1/4), \qquad H_3(n,n) \sim \frac{64^N}{16\pi N^3} Z_3(1/4), \tag{17}$$

where, for $0 < q < 1$,

$$Z_r(q) = \sum_{0 \le b_1 < \cdots < b_r} \prod_{i<j} (b_j - b_i)^2 q^{b_1 + \cdots + b_r}.$$

This argument explicitly bounds the whole tail, rather than interchanging a pointwise limit and an unbounded sum.

Put $\mu_k(q) = \sum_{b \ge 0} b^k q^b$, with $b^0 = 1$ also for $b = 0$. Equivalently,

$$\mu_k(q) = \left(q \frac{d}{dq} \right)^k \frac{1}{1-q}.$$

Cauchy–Binet for the weighted Vandermonde matrix, first with bounded column indices and then in the limit, gives

$$Z_r(q) = \det[\mu_{i+j}(q)]_{0 \le i,j < r}.$$

Every moment converges for $0 < q < 1$, and a finite determinant is a polynomial in its entries, which justifies that limit. Direct differentiation and the two small determinants give

$$Z_2(q) = \frac{q}{(1-q)^4}, \qquad Z_3(q) = \frac{4q^3}{(1-q)^9}.$$

In particular,

$$Z_2(1/4) = \frac{64}{81}, \qquad Z_3(1/4) = \frac{16384}{19683}.$$

Finally $N = n - 2$ and $N/n \to 1$, while

$$\frac{64/81}{16^2} = \frac{1}{324}, \qquad \frac{16384/19683}{16 \cdot 64^2} = \frac{1}{78732}.$$

Substitution into (17) proves Theorem 1. □

6 Reproducible finite checks

The accompanying script `hardinian.py` evaluates the exact integer formulas through $n = 201$. It also enumerates arrays directly from the definition for $1 \le n \le 6$ and evaluates the complementary minors independently for $3 \le n \le 8$, in both cases for $r = 2, 3$. All comparisons pass. The computed values agree with the 37 supplied terms of OEIS A253217 and the first 24 supplied terms of A252998, together with its entries at $n = 100, 101$ [2, 3]. The script includes exact symbolic checks of Z_2 and Z_3 when SymPy is installed. The accompanying result file records an actual successful run. These checks help detect indexing and constant errors; the limits for unbounded n are proved in Sections 4–5.

AI assistance and verification status. This research was completed with AI assistance. The mathematical argument, source references, and reproducible checks are provided for evaluation. This version does not claim proof-assistant verification or external peer review.

References

[1] R. Dougherty-Bliss and M. Kauers, *Hardinian Arrays*, Electronic Journal of Combinatorics **31**(2) (2024), P2.9. doi:10.37236/12358.

[2] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A253217; table of values. Accessed October 13, 2026.

[3] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A252998; table of values. Accessed October 13, 2026.